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CODE & DATA
github.com/memess2001/lenia-search

28 kernels	16 growth functions	291 / 400 behavioral niches	50 moving solitons	9 / 9 robust trials
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Research Question

Standard Lenia

$$A_{t+1} = clip(A_t + dt \cdot G(K * A_t), 0, 1)$$

Rich lifelike behavior but historically confined to Gaussian kernels + bell-shaped growth functions.
What other mathematics can produce life?

Expanded Search Space

1 Function families

4 → 28 kernels · 4 → 16 growth functions

2 Geometric substrates

flat $\mathbb{R}^2$ · sphere S^2 · torus T^2 · hyperbolic $\mathbb{H}^2$

3 Open expressions

LLM-generated PyTorch rules using conv, ∇^2 , gradient, sin, cos, tanh, exp

Search & Evaluation Protocol

Track	Res.	Steps	Evals	Seeded
A (parametric)	128 ²	8,000	~125K	287
B (LLM)	64 ²	2,000	~5K	287
C (random)	64 ²	2,000	1,000 (809 valid)	None
Motion	256 ²	10,000	~15K	None

Controlled comparison

Key entries re-evaluated at 128², 8,000 steps, 10 seeds.

Unequal track budgets are reported explicitly.

Expanded Families Delivered

11 / 13 newly added kernels produced archive entries

Morse		7
sech		7
Wendland C2		6
Riesz		6

Four geometries contributed comparably (≈21–29% each).

MAP-Elites Archive

20 × 20 grid = 400 niches

Complexity: LZMA ratio of the final field quantized to 8 bits.

Alive fraction: fraction of cells with A > 0.01.

Elite rule: retain the highest-complexity entry per niche; all tracks compete for the same cells.

Coverage measures diversity illumination, not independent quality optimization.

Central Finding

Pattern morphology: organic (Track A) vs. geometric (Track B)

Track A #1
sech / FitzHugh-Nagumo
controlled mean = 0.998

Track A #4
Riesz / Gaussian bell
controlled mean = 0.951

Track A #3
Yukawa / Laplace peak
controlled mean = 0.960

Track B #6 (hero)
cos(4π∇²A)
controlled mean = 0.939

The Hero Discovery

growth center modulated by cos(4π · ∇²A)

1 Not expressible by single-kernel Lenia: admissible kernels are non-negative and normalized; ∇² is zero-mean and sign-indefinite.

2 Not expressible by additive multi-kernel Lenia: ΣG_i(K_i*A) cannot reproduce this nonlinear coupling.

3 Scope: not a new PDE or universal impossibility. Multi-channel Lenia / NCA can represent it; novelty is reachability beyond standard parametric families.

Why the full expression matters

1 Laplacian-modulated center

2 band-pass density window

3 harmonic modulation

crystallographic rings
complexity ≈ 0.94

Center modulation alone → Turing-like spots, complexity ≈ 0.13.

Hero robustness across resolutions

Resolution	Alive frac.	Complexity	Clusters	Entropy
128 ² (n=10)	0.977±0.001	0.939±0.002	6.1±1.6	3.49±0.11
256 ² (n=10)	0.983±0.001	0.899±0.002	29.3±2.7	3.04±0.08

20 runs at two resolutions; coefficient of variation < 0.3%.

Unified archive coverage

Final archive · 291/400 bins

Alive fraction bin

Complexity bin

Track A (parametric) Track B (LLM) Empty

★ Hero entry · controlled complexity 0.939

291 / 400
niches filled · 72.8%

285 A · 6 B
archive contributions

Rank #6
controlled · structurally novel

Locomotion Evidence

bessel / gaussian_bell — 9/9 robust · direction std = 2.1°

t = 0

t = 2,000

t = 4,000

t = 6,000

t = 8,000

t = 10,000

periodic boundary recentered · texture → stationary → moving soliton

Six Moving Soliton Morphologies

morse / relu_like
tadpole · d=75

sinc / step_function
ring · d=47

elliptical / sigmoid_pair
crescent · d=55

rbf_mixture / laplace_peak
pac-man · d=67

polynomial / sigmoid_pair
ring · d=38

bessel / gaussian_bell
ring · 9/9

50 solitons · 17 combinations · 49/50 from expanded families

Locomotion Search & Candidates

$$K \leftarrow K \cdot [1 + \epsilon \cos(\theta - \theta_0)] \quad \epsilon \in [0.1, 0.6]$$

Fitness = concentration × displacement
Gates: d>20 px · c>0.7 · alive<15%
Injected anisotropy (not emergent); wider basins appear under coarse search.

Kernel family	Candidates	Best d
sinc	6	173 px
riesz	3	119 px
elliptical	3	93 px
morse	1	75 px
yukawa	14	56 px
lennard_jones	4	55 px
polynomial*	1	38 px

*Original Lenia kernel paired with a new growth function.

Controlled Track Comparison

Metric	A (parametric)	B (LLM)	C (random)
Evaluations	~125K	~5K	1,000 (809 valid)
Archive bins	291	6	136
Best fitness	0.997	0.939*	0.967
Entries >0.9	12	1	3

Function space → diversity: 1,000 random expressions fill 136 bins.

Parametric bias → quality: Track A wins 132/136 comparisons.

LLM → tuning: Track C finds cos(∇²A) but misses 4π resonance.
*Track B: 128², 8,000 steps, 10 seeds.

LIMITATIONS

- 1 Lenia substrate only · 2 Unequal track budgets · 3 Track B warm-started
- 4 High LLM-expression failure rate · 5 Complexity doubles as descriptor and fitness

NEXT EXPERIMENTS

- Cross-substrate transfer · 512²+ richer creatures
- Matched-budget comparison